

in the forward fuselage of the Orbiter vehicle. Payloads are carried in the mid-fuselage payload bay, and the Orbiter's main engines and maneuvering thrusters are located in the aft fuselage.

Forward Fuselage

The cockpit, living quarters and experiment operator's station are located in the forward fuselage. This area houses the pressurized crew module and provides support for the nose section, the nose gear and the nose gear wheel well and doors.

Crew Module

The 65.8-cubic-metre crew station module is a three-section pressurised working, living and stowage compartment in the forward portion of the Orbiter. It consists of the flight deck, the middeck/equipment bay and an airlock. Outside the aft bulkhead of the crew module in the payload bay, a docking module and a transfer tunnel with an adapter can be fitted to allow crew and equipment transfer for docking and extravehicular operations. The two-level crew module has a forward flight deck with the commander's seat positioned on the left and the pilot's seat on the right.

The flight deck is designed in the usual pilot/co-pilot arrangement, which permits the vehicle to be piloted from either seat and permits one-man emergency return. Each seat has manual flight controls, including rotation and translation hand controllers, rudder pedals and speed-brake controllers. The flight deck seats four. The on-orbit displays and controls are at the aft end of the flight deck/crew compartment. The displays and controls on the left are for operating the Orbiter, and those on the right are for operating and handling the payloads. More than 2,020 separate displays and controls are located on the flight deck.

Six pressure windshields, two overhead windows and two rear-viewing payload bay windows are located in the upper flight deck of the crew module, and a window is located in the crew entrance/exit hatch located in the midsection, or deck, of the crew module.

The middeck contains provisions and stowage facilities for four crew sleep stations. Stowage for the lithium hydroxide canisters and other gear, the waste management system, the personal hygiene station and the work/dining table is also provided in the middeck.

The nominal maximum crew size is seven. The middeck can be reconfigured by adding three rescue seats in place of the modular stowage and sleeping provisions. The seating capacity will then accommodate the rescue flight crew of three and a maximum rescued crew of seven.